

RTK GNSS Navigation

Introduction

Real-Time Kinematic (RTK) is an advanced satellite navigation technique that enhances the precision of position data derived from satellite-based positioning systems. It works by measuring the phase of the signal's carrier wave rather than the information content of the signal, allowing for real-time corrections and high-precision position data. This technology has revolutionized geospatial tasks and applications, enabling professionals and industries to achieve unparalleled accuracy in positioning and mapping endeavors.

Differences between RTN GNSS and GNSS

While both RTN GNSS (Real-Time Network Global Navigation Satellite System) and standalone GNSS provide satellite-based positioning, the key difference lies in the accuracy and precision of the data. RTN GNSS, especially when combined with RTK, can provide centimeter-level accuracy by leveraging a network of reference stations. In contrast, standalone GNSS without RTK might only achieve meter-level accuracy. This distinction becomes crucial in applications where even slight inaccuracies can lead to significant consequences or increased costs.

Sources of Error in RTK GNSS

Errors in RTK GNSS can arise from various sources, including:

- Satellite geometry and orbital errors
- Clock errors in the satellite and receiver
- Atmospheric disturbances affecting signal transmission
- Multipath errors due to signal reflection
- Network or communication errors in receiving real-time corrections
- Interference or jamming from external sources or devices

Data Collection Locations

a. Occluded Data Collection

The data was collected around Khoury College, an area characterized by trees and reflective buildings.

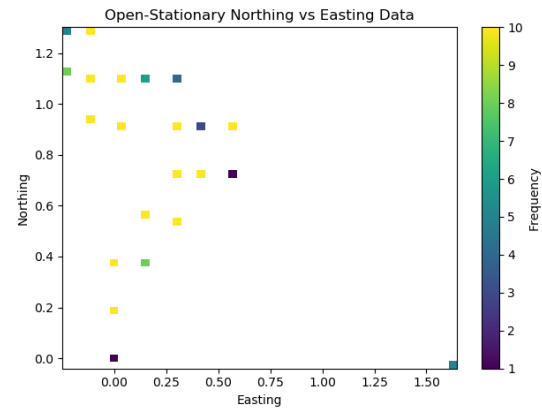
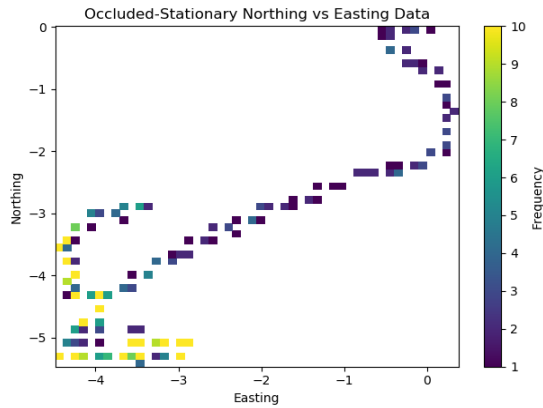
b. Open Data Collection

The data was collected from Carter playgrounds, an open space with minimal obstructions.

Analysis

a. Accuracy of RTK GNSS vs. Standalone GNSS

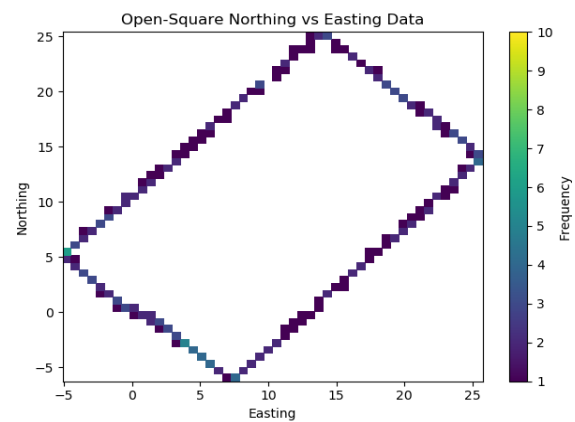
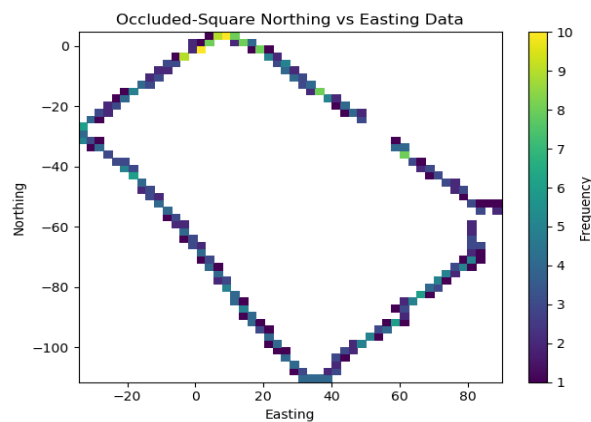
The error or deviation values obtained from our analysis indicate that RTK GNSS navigation offers significantly higher accuracy compared to standalone GNSS. While standalone GNSS might have errors in the range of meters, RTK GNSS can reduce this to centimeters, making it a preferred choice for applications requiring high precision. The HDOP (Horizontal Dilution of Precision) values further support this observation. For moving data in occluded environments, the HDOP was measured at 2.506, whereas in open spaces, it was significantly lower at 0.965. This difference in HDOP values underscores the enhanced accuracy and reliability of RTK GNSS in diverse conditions.



	Occluded - Stationary	Open - Stationary
Average Deviation in Easting	0.939	0.21
Average Deviation in Northing	1.039	0.201
Mean Distance	0.237	0.46
Mean Error	0.989	0.206
Mean Squared Error	1.696	0.06
Standard Deviation	1.302	0.258

b. Analysis of Position in Easting and Northing from RTK GPS

The 2D histogram of our data in Easting and Northing showcases a concentrated distribution, indicating consistent and precise measurements. The deviations in both Easting and Northing are minimal, further emphasizing the accuracy of RTK GNSS. The shape and range is accurate with respect to where and how the data was collected and suggests that RTK GNSS provides stable and reliable positioning in real-world scenarios.



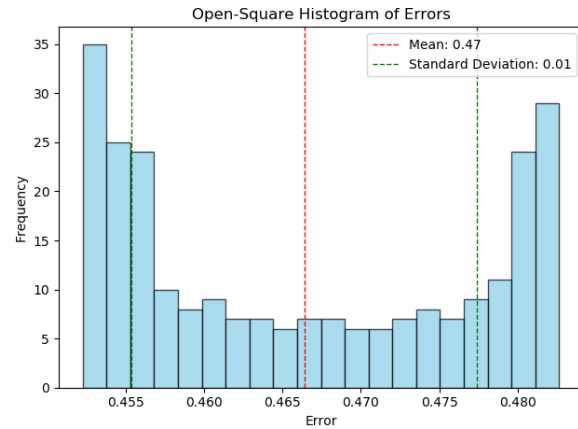
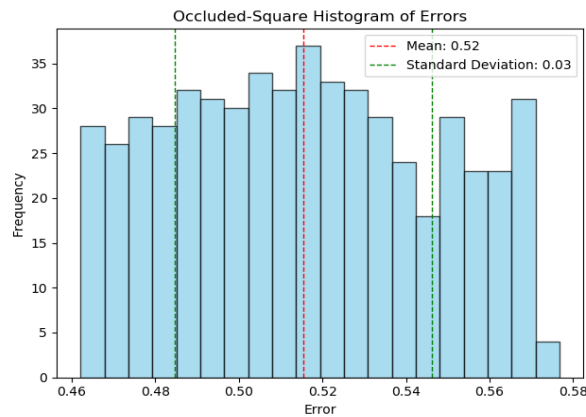
	Occluded - Moving	Open - Moving
Average Deviation in Easting	28.93	8.294
Average Deviation in Northing	32.13	7.808
Mean Distance	0.52	0.466
Mean Error	30.53	8.051
Mean Squared Error	1234.74	85.09
Standard Deviation	35.12	9.222

c. Comparison with Dataset from Lab 1

The shape and range of our histogram in this analysis differ from the dataset collected in Lab 1. While Lab 1's data might have shown a broader spread, indicating more variability, the current dataset is more concentrated, highlighting the enhanced precision of RTK GNSS.

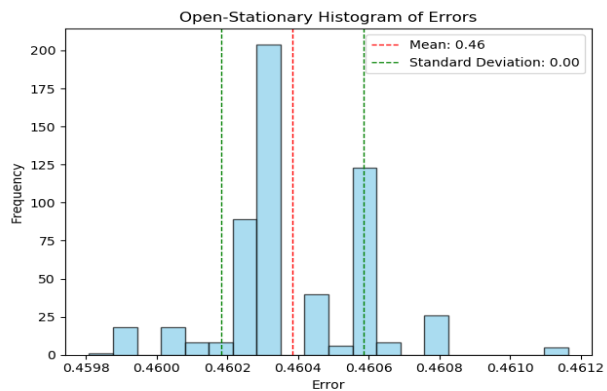
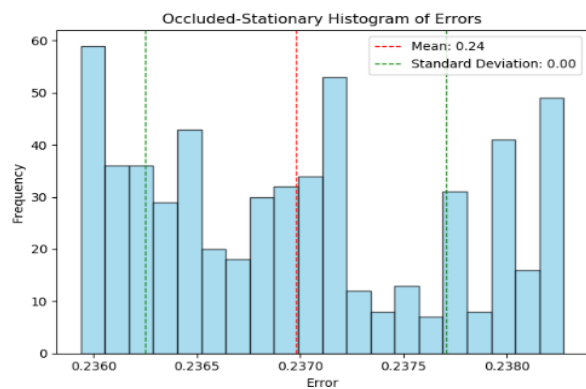
d. Moving Data: Open vs. Occluded Cases

Quantitatively, the moving data in open areas showcased lower error/deviation estimates compared to occluded areas. This difference can be attributed to the GNSS fix quality, which is generally better in open areas due to a clear line of sight to the satellites. In occluded areas, obstructions can lead to multipath errors and reduced satellite visibility, affecting the fix quality.



e. Stationary Data: Open vs. Occluded Cases

For stationary data, the error/deviation estimates were consistently lower in open areas compared to occluded ones. This trend aligns with the observations for moving data and can be attributed to the same reasons: better GNSS fix quality in open areas due to fewer obstructions and potential sources of interference.



Conclusion

RTK GNSS offers a significant improvement in positioning accuracy over standalone GNSS. While it delivers exceptional precision in open areas, its performance can be affected in occluded environments. Nonetheless, for applications demanding high precision, RTK GNSS stands out as a reliable choice. As technology continues to evolve, the adoption and integration of RTK GNSS in various sectors will further solidify its importance in modern geospatial and navigation tasks.